Software Requirements Specification (SRS)

Project: Event Management System (EMS)

Version: 2.0

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Course: UE24CS341A Software Engineering, PES University. Group 7 (to confirm)

Revision history

Version	Date	Author	Change summary	Approval
1.0	11-10-2026	Team	Initial SRS with requirements, security section and use case diagrams	
2.0	10-11-2026	Team	Final version after Sprint 2: requirements unchanged; traceability matrix completed with design, modules, test cases and test status; implementation deviations listed in section 8.1	

Approvals

Role	Name	Signature / Email	Date
Course faculty			
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1. Introduction

1.1 Purpose

This document is the Software Requirements Specification (SRS) for the Event Management System (EMS). It defines the functional and non-functional requirements, external interfaces, security requirements and verification criteria of the system. It is the baseline for the architecture and design specification, the test plan, the project plan and the implementation.

1.2 Scope

EMS is a web application that lets organizers create and manage events and lets attendees discover events and register for them.

In scope:

- User accounts with role-based access (Attendee, Organizer, Admin)
- Event creation, editing, publishing, cancellation, categorization, search and filtering
- Event registration, waitlisting, cancellation and ticket generation
- Attendance check-in, email and in-app notifications, event reports and CSV export
- Administrative user management and a platform dashboard

Out of scope:

- Online payments and paid ticketing
- Venue booking and seat maps
- Native mobile applications
- Integration with third-party calendars or social networks

1.3 Audience

Developers, QA engineers, project guides and evaluators, and the maintenance team.

1.4 Definitions, Acronyms and Abbreviations

Term	Meaning
EMS	Event Management System
SRS / SAD / STP / RTM	Software Requirements Specification / Software Architecture and Design Specification / Software Test Plan / Requirements Traceability Matrix
RBAC	Role-Based Access Control
JWT	JSON Web Token, used for session authentication
API / REST	Application Programming Interface / Representational State Transfer
HTTPS / TLS	Hypertext Transfer Protocol Secure / Transport Layer Security
SMTP	Simple Mail Transfer Protocol, used for sending email
QR code	Quick Response code, a 2D barcode that encodes a ticket ID
CSV	Comma-Separated Values file
FIFO	First In, First Out
Waitlist	Ordered queue of attendees waiting for a seat in a full event

2. Overall Description

2.1 Product Perspective

EMS is a standalone client-server web application. A React.js single-page frontend communicates with a Node.js/Express REST API, which stores data in a PostgreSQL database and sends email through an external SMTP service.

2.2 Major Product Functions

- **F1 Authentication and role management:** account registration, login, logout, role-based access, profile management, user administration
- **F2 Event creation and management:** create, edit, publish, cancel, categorize, browse, search and filter events
- **F3 Event registration and ticketing:** register, waitlist, cancel, automatic waitlist promotion, ticket with QR code, attendee lists
- **F4 Check-in, notifications and reports:** ticket check-in, email and in-app notifications, event reports, CSV export, admin dashboard

2.3 User Roles and Characteristics

Role	Description	Characteristics
Visitor	Unauthenticated user	Can browse and search published events and create an account
Attendee	Registered user who attends events	Basic web skills, expects quick registration and a clear confirmation
Organizer	Registered user who runs events	Moderate skills, needs control over event details, capacity and attendance data
Admin	Platform administrator	Manages users, roles and categories, monitors platform activity
Email Service	External system (secondary actor)	Delivers notification emails on behalf of EMS

2.4 Operating Environment

- Client: latest two versions of Google Chrome, Mozilla Firefox and Microsoft Edge, on desktop and mobile screens from 360 px to 1920 px width
- Server: Node.js LTS on Linux, packaged with Docker
- Database: PostgreSQL

2.5 Constraints

- Technology stack: React.js, Node.js with Express, PostgreSQL
- Development follows Agile with GitHub Projects, GitHub Actions for CI and SonarQube for static analysis
- No payment processing, so every event is free to register for
- The project must be completed within the academic semester by a team of four

2.6 Assumptions and Dependencies

- Users have an internet connection and a valid email address
- An SMTP service is available for notification emails

- All event times are handled in a single configured time zone
- A device camera is optional. Check-in also works by typing the ticket ID.

2.7 Requirement ID Conventions

Prefix	Meaning	Range in use
EVT-F-###	Functional requirement	F1: 001-007, F2: 010-016, F3: 020-027, F4: 030-037
EVT-NF-###	Non-functional requirement	001-006
EVT-SR-###	Security requirement	001-007

3. External Interface Requirements

3.1 User Interfaces

- Responsive web interface with a navigation bar that depends on the user role
- Pages: home and event listing, event details, login and sign-up, my registrations, organizer dashboard, event form, check-in screen, reports, admin console
- Accessibility: keyboard navigation, text alternatives for images and sufficient colour contrast

3.2 Hardware Interfaces

EMS needs no special hardware. Optional: a device camera for scanning QR codes at check-in.

3.3 Software Interfaces

Interface	Purpose
EMS REST API (JSON over HTTPS)	Communication between the React frontend and the Node.js backend
PostgreSQL	Persistent storage of users, events, registrations, notifications and audit logs
SMTP email service	Sending confirmation, waitlist, cancellation and reminder emails

3.4 Communications Interfaces

- All client-server traffic uses HTTPS (TLS 1.2 or higher)
- The API exchanges JSON and authenticates requests with a JWT in the Authorization header
- Email is sent asynchronously so a slow mail server does not block user actions

The requirements meet the minimum set for the project: 30 functional requirements, 6 non-functional requirements, 3 security objectives and 7 security requirements.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow EVT-F-###. Each feature is owned end to end by one team member.

4.1 Authentication and Role Management (Feature F1)

Feature owner: Varun Rao (PES1UG24CS520)

Description: Allows visitors to create accounts and log in, enforces role-based access for Attendees, Organizers and Admins, and lets users and admins manage accounts.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
EVT-F-001	The system shall allow a visitor to register an account by providing name, email, password and role (Attendee or Organizer).	Functional	High	Visitor	AC-EVT-F-001: A valid form creates the account, and an existing email is rejected with the message "Email already registered". Test: TC-AUTH-01, TC-AUTH-02	Password must have at least 8 characters with a letter and a digit
EVT-F-002	The system shall authenticate a user with email and password and issue a session token valid for 60 minutes.	Functional	High	All users	AC-EVT-F-002: Correct credentials return a token and open the role dashboard, and wrong credentials return an error without revealing which field was wrong. Test: TC-AUTH-03	Depends on EVT-F-001 and EVT-SR-002
EVT-F-003	The system shall lock an account for 15 minutes after 5 consecutive failed login attempts.	Functional	High	Security	AC-EVT-F-003: The 6th attempt within the lock period is refused even with the correct password, and login works again after 15 minutes. Test: TC-AUTH-04	Failed attempts are recorded in the audit log (EVT-SR-006)
EVT-F-004	The system shall allow a logged-in user to log out, which invalidates the session.	Functional	Medium	All users	AC-EVT-F-004: After logout the old token is rejected with HTTP 401. Test: TC-AUTH-05	
EVT-F-005	The system shall restrict every feature to the roles permitted to use it and respond with HTTP 403 when a user lacks permission.	Functional	High	Security	AC-EVT-F-005: An Attendee calling an Organizer-only or Admin-only action receives HTTP 403 and no data change occurs. Test: TC-AUTH-06	Applies to all features
EVT-F-006	The system shall allow an Admin to view all users, change a user's role, and deactivate or reactivate an account.	Functional	Medium	Admin	AC-EVT-F-006: A deactivated user cannot log in, and a role change takes effect on the user's next login. Test: TC-AUTH-07	An Admin cannot deactivate their own account
EVT-F-007	The system shall allow a user to view and update their name and phone number and to change their password by supplying the current password.	Functional	Low	All users	AC-EVT-F-007: A correct current password allows the change, and a wrong one is rejected. Test: TC-AUTH-08	

4.2 Event Creation and Management (Feature F2)

Feature owner: Thanishq Raam Patil (PES1UG24CS502)

Description: Lets organizers create, edit, publish and cancel events, lets admins maintain event categories, and lets all users browse, search and filter published events.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
EVT-F-010	The system shall allow an Organizer to create an event with title, description, category, venue, start date-time, end date-time and capacity.	Functional	High	Organizer	AC-EVT-F-010: Submitting all mandatory fields saves the event with status Draft and shows it in the organizer's event list. Test: TC-EVT-01	Requires Organizer role (EVT-F-005)
EVT-F-011	The system shall validate event data and reject an event	Functional	High	Organizer	AC-EVT-F-011: Each invalid case returns a specific error message	Also enforces EVT-SR-003

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
	whose title is empty, whose start is not in the future, whose end is not after the start, or whose capacity is not a whole number from 1 to 10000.				and nothing is saved. Test: TC-EVT-02, TC-EVT-03	
EVT-F-012	The system shall allow an Organizer to edit only their own events, and shall refuse to reduce the capacity below the number of confirmed registrations.	Functional	High	Organizer	AC-EVT-F-012: Edits to own events are saved, a capacity below confirmed registrations is rejected, and editing another organizer's event gives HTTP 403. Test: TC-EVT-04	Depends on F3 registration count
EVT-F-013	The system shall allow an Organizer to publish a Draft event, after which it is visible to all users, and shall keep Draft events hidden from Attendees and Visitors.	Functional	High	Organizer	AC-EVT-F-013: A published event appears in public search, and a Draft does not. Test: TC-EVT-05	
EVT-F-014	The system shall allow an Organizer to cancel their own event, set its status to Cancelled and cancel all its registrations.	Functional	High	Organizer	AC-EVT-F-014: The event shows as Cancelled, every registration for it becomes Cancelled, and cancellation notifications are triggered (EVT-F-032). Test: TC-EVT-06	Depends on EVT-F-032
EVT-F-015	The system shall allow an Admin to add, rename and delete event categories, and shall refuse to delete a category that is in use.	Functional	Medium	Admin	AC-EVT-F-015: A new category is selectable on the event form, and deleting a category with events is refused with a message. Test: TC-EVT-07	
EVT-F-016	The system shall allow any user to browse published events, search by keyword in title or description, and filter by category and date range, with results paginated at 10 per page.	Functional	High	Visitor, Attendee	AC-EVT-F-016: A keyword returns only matching published events, filters combine correctly, and page 2 shows the next 10 results. Test: TC-EVT-08, TC-EVT-09	Cancelled events are shown with a Cancelled label

4.3 Event Registration and Ticketing (Feature F3)

Feature owner: Vishruta (PES1UG24CS538)

Description: Lets attendees register for published events, joins them to a waitlist when an event is full, handles cancellations with automatic waitlist promotion, and provides tickets with QR codes.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
EVT-F-020	The system shall allow a logged-in Attendee to register for a published event that has not started and has a free seat, and shall create a confirmed registration with a unique ticket ID.	Functional	High	Attendee	AC-EVT-F-020: The registration is Confirmed, the seat count drops by 1, and a unique ticket ID is shown. Registering for a Draft, Cancelled or started event is refused. Test: TC-REG-01, TC-REG-02	Depends on EVT-F-013
EVT-F-	The system shall prevent an	Functional	High	Organize	AC-EVT-F-021: A second	

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
021	Attendee from holding more than one active registration (Confirmed or Waitlisted) for the same event.			r	registration attempt is rejected with the message "Already registered". Test: TC-REG-03	
EVT-F-022	The system shall add an Attendee to a first-in-first-out waitlist when the event has no free seats.	Functional	High	Attendee, Organizer	AC-EVT-F-022: The registration status is Waitlisted and the waitlist position is shown. Test: TC-REG-04	
EVT-F-023	The system shall allow an Attendee to cancel their own registration before the event starts and shall release the seat.	Functional	High	Attendee	AC-EVT-F-023: The status becomes Cancelled and the free seat count increases by 1. Cancelling after the start time is refused. Test: TC-REG-05	
EVT-F-024	The system shall automatically promote the first Waitlisted Attendee to Confirmed when a seat is released and shall notify them.	Functional	High	Attendee	AC-EVT-F-024: The first waitlisted attendee becomes Confirmed, the next one moves up, and a notification is triggered (EVT-F-032). Test: TC-REG-06	Depends on EVT-F-032
EVT-F-025	The system shall show an Attendee a list of their registrations with event name, date and status (Confirmed, Waitlisted, Cancelled).	Functional	Medium	Attendee	AC-EVT-F-025: The list shows only the logged-in attendee's registrations, each with the correct status. Test: TC-REG-07	See EVT-SR-005
EVT-F-026	The system shall provide for each Confirmed registration a ticket page showing the ticket ID, event details and a QR code that encodes the ticket ID.	Functional	Medium	Attendee	AC-EVT-F-026: Scanning the QR code returns the ticket ID shown on the page. Test: TC-REG-08	Used by check-in (EVT-F-030)
EVT-F-027	The system shall allow an Organizer to view the list of registrations (Confirmed, Waitlisted, Cancelled) for their own events.	Functional	Medium	Organizer	AC-EVT-F-027: The list shows attendee name, email, status and registration time, and is refused for another organizer's event. Test: TC-REG-09	

4.4 Attendance Check-in, Notifications and Reports (Feature F4)

Feature owner: Nithin Kumar M (PES1UG25CS829)

Description: Lets organizers check attendees in with their tickets, sends email and in-app notifications for key events, and provides attendance reports, CSV export and an admin dashboard.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
EVT-F-030	The system shall allow an Organizer to check in an attendee for their own event by scanning or entering the ticket ID, and shall record the check-in time.	Functional	High	Organizer	AC-EVT-F-030: A valid Confirmed ticket becomes Checked-in with a timestamp and the screen shows the attendee name. Test: TC-CHK-01	Uses the ticket from EVT-F-026
EVT-F-031	The system shall reject a check-in for a ticket that is	Functional	High	Organizer	AC-EVT-F-031: Each of the five cases shows its own message	

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
	invalid, Cancelled, Waitlisted, already used, or belongs to a different event, with a specific message for each case.				and no check-in is recorded. Test: TC-CHK-02, TC-CHK-03	
EVT-F-032	The system shall send an email to the attendee when their registration is confirmed, when they are promoted from the waitlist, when they cancel, and when the event is cancelled.	Functional	High	Attendee	AC-EVT-F-032: Each of the four situations sends one email with the event name and the new status. Test: TC-CHK-04	Triggered by EVT-F-014, EVT-F-020, EVT-F-023 and EVT-F-024
EVT-F-033	The system shall send a reminder email 24 hours before the start of an event to every Confirmed attendee.	Functional	Medium	Attendee	AC-EVT-F-033: Each Confirmed attendee gets exactly one reminder 24 hours (plus or minus 5 minutes) before the start. Test: TC-CHK-05	Uses a scheduled job
EVT-F-034	The system shall show each user an in-app notification list that records the notifications sent to them and lets them mark notifications as read.	Functional	Low	Attendee	AC-EVT-F-034: New notifications appear as unread and change to read when marked. Test: TC-CHK-06	
EVT-F-035	The system shall give an Organizer a report for each of their events showing counts of Confirmed, Waitlisted, Cancelled and Checked-in registrations and the attendance rate (checked-in divided by confirmed, as a percentage).	Functional	High	Organizer	AC-EVT-F-035: For 10 confirmed registrations with 8 checked in, the attendance rate shows 80%. Test: TC-CHK-07	
EVT-F-036	The system shall allow an Organizer to export the attendee list of their own event as a CSV file with name, email, status and check-in time.	Functional	Medium	Organizer	AC-EVT-F-036: The downloaded CSV opens correctly and has one row per registration with the four columns. Test: TC-CHK-08	
EVT-F-037	The system shall give an Admin a dashboard showing total users by role, total events by status and total registrations.	Functional	Low	Admin	AC-EVT-F-037: The dashboard totals match the database counts. Test: TC-CHK-09	

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs EVT-NF-###.

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
EVT-NF-001	At least 95% of API requests shall complete within 2 seconds with 100 concurrent users on the reference test environment.	Performance	High	Load test report shows the 95th percentile at or below 2 s. Test: TC-NFR-01
EVT-NF-002	The system shall be available at least 99% of each month, measured by health-check pings every minute, excluding announced	Reliability	Medium	Uptime report shows 99% or higher. Test: TC-NFR-02

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
	maintenance.			
EVT-NF-003	A new Attendee shall be able to register for an event in no more than 3 clicks after opening its details page, and every page shall display correctly at screen widths from 360 px to 1920 px.	Usability	Medium	Usability walkthrough with 5 users and layout checks at 360 px and 1920 px. Test: TC-NFR-03
EVT-NF-004	Under 500 simultaneous registration attempts for an event, the system shall never confirm more registrations than the event capacity.	Scalability / Correctness	High	Concurrency test finds confirmed registrations less than or equal to capacity in every run. Test: TC-NFR-04
EVT-NF-005	The backend code shall reach at least 70% unit test line coverage and pass the static-analysis quality gate with no blocker or critical issues before each release.	Maintainability	Medium	Coverage report shows 70% or higher and the SonarQube quality gate passes. Test: TC-NFR-05
EVT-NF-006	The web interface shall work correctly on the latest two versions of Google Chrome, Mozilla Firefox and Microsoft Edge.	Compatibility	Medium	All acceptance tests pass on each listed browser. Test: TC-NFR-06

5.1 Security

5.1.1 Security Objectives

The following security objectives apply to the whole Event Management System. The security requirements in section 5.1.2 (EVT-SR-001 to EVT-SR-007) implement them.

ID	Objective	Description	Related Requirements
SO-1	Confidentiality	Credentials and personal data of attendees, organizers and admins are protected from unauthorized access, in storage and in transit.	EVT-SR-001, EVT-SR-004, EVT-SR-005
SO-2	Integrity	Event, registration and attendance data can be created or changed only by authenticated users who hold the right role and own the data, and inputs cannot be used to corrupt the system.	EVT-SR-002, EVT-SR-003, EVT-SR-005
SO-3	Accountability and Availability	Security-relevant actions are recorded so they can be traced to a user, and abuse such as brute-force or request flooding is limited so the service stays available to genuine users.	EVT-SR-006, EVT-SR-007

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria / Test Case Ref
EVT-SR-001	Passwords shall be stored only as salted bcrypt hashes (cost factor 10 or higher) and shall never appear in logs or API responses.	Security	High	Database inspection shows no plaintext password, and log review finds none. Test: TC-SEC-01
EVT-SR-002	Session tokens shall be signed JWTs that expire after 60 minutes, and expired, tampered or missing tokens shall be rejected with HTTP 401.	Security	High	A modified or expired token gets HTTP 401. Test: TC-SEC-02
EVT-SR-003	All user input shall be validated and sanitized on the server, and database access shall use parameterized queries, to prevent SQL injection	Security	High	Injection strings such as `' OR 1=1 --` and `<script>` in any field are rejected or stored harmlessly. Test:

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria / Test Case Ref
	and cross-site scripting.			TC-SEC-03
EVT-SR-004	All communication between the browser and the server shall use HTTPS with TLS 1.2 or higher, and plain HTTP requests shall be redirected or refused.	Security	High	A TLS scan shows no protocol below 1.2 and an HTTP request is redirected. Test: TC-SEC-04
EVT-SR-005	The system shall enforce ownership checks so that an Attendee can access only their own registrations and tickets, and an Organizer only the registrations of their own events.	Security	High	Requesting another user's registration or ticket by ID returns HTTP 403. Test: TC-SEC-05
EVT-SR-006	The system shall record an audit log entry with timestamp, user ID and action for every login success or failure, role change, event cancellation and check-in, and the log shall never contain passwords or tokens.	Security	High	Performing each listed action creates one correct log entry, and a log search finds no passwords or tokens. Test: TC-SEC-06
EVT-SR-007	The system shall limit each IP address to 100 API requests per minute and to 10 login attempts per minute, and shall answer excess requests with HTTP 429.	Security	Medium	The 101st request in a minute returns HTTP 429. Test: TC-SEC-07

6. Quality Attributes and Acceptance Tests

6.1 Exit Criteria for Acceptance

- All High-priority functional requirements are implemented and verified
- No critical or high-severity defect is open
- All non-functional and security requirements meet their measured targets
- The Requirements Traceability Matrix (section 8) shows every requirement mapped to at least one passed test case

6.2 Acceptance Test Suites

Suite	Covers	Test Case Prefix
Authentication and Roles	EVT-F-001 to EVT-F-007	TC-AUTH
Event Management	EVT-F-010 to EVT-F-016	TC-EVT
Registration and Ticketing	EVT-F-020 to EVT-F-027	TC-REG
Check-in, Notifications and Reports	EVT-F-030 to EVT-F-037	TC-CHK
Non-Functional (performance, availability, usability, concurrency, maintainability, compatibility)	EVT-NF-001 to EVT-NF-006	TC-NFR
Security	EVT-SR-001 to EVT-SR-007	TC-SEC

6.3 Quality Attributes Summary

Attribute	Target
Performance	95% of API requests within 2 s at 100 concurrent users
Reliability	99% monthly availability
Usability	A new attendee registers for an event in 3 clicks or fewer after opening it
Scalability and correctness	No overbooking under 500 simultaneous registration attempts
Maintainability	At least 70% unit test line coverage and a passing static-analysis quality gate
Compatibility	Latest two versions of Chrome, Firefox and Edge

7. System models and diagrams

7.1 UML Use-Case diagram

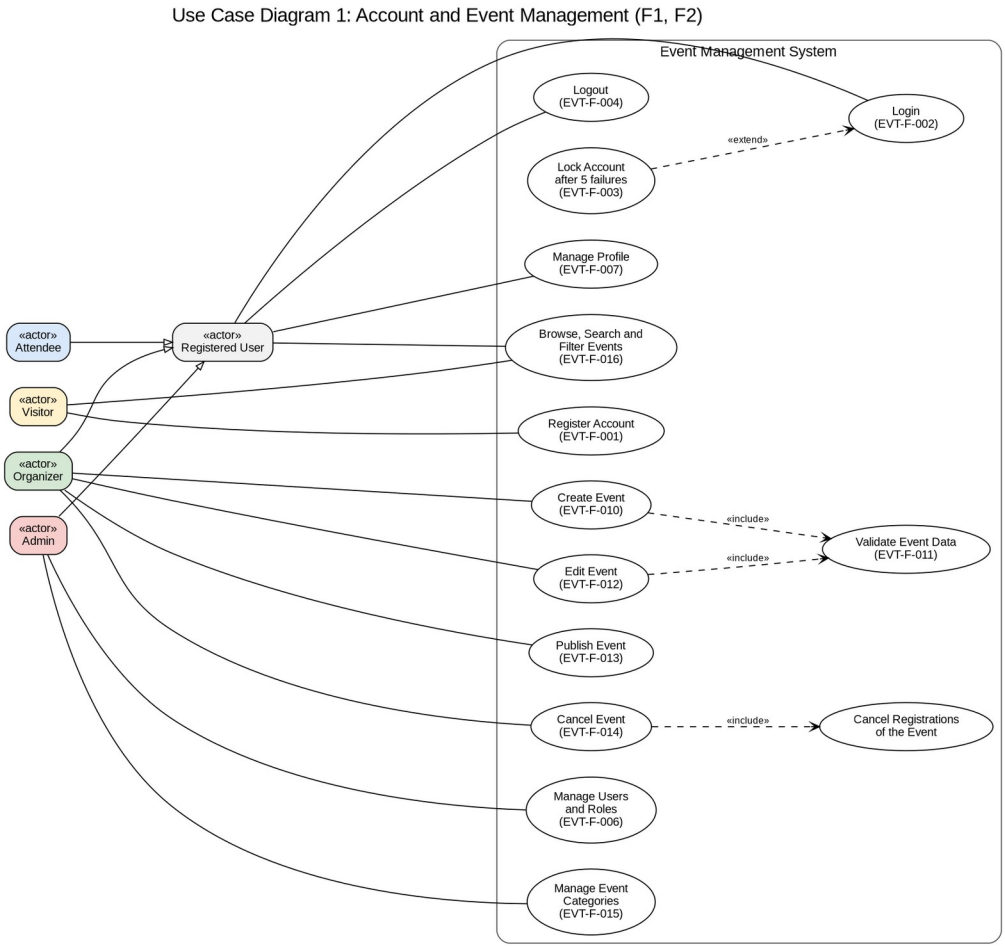


Figure 7.1: Use case diagram 1, account and event management (F1, F2)

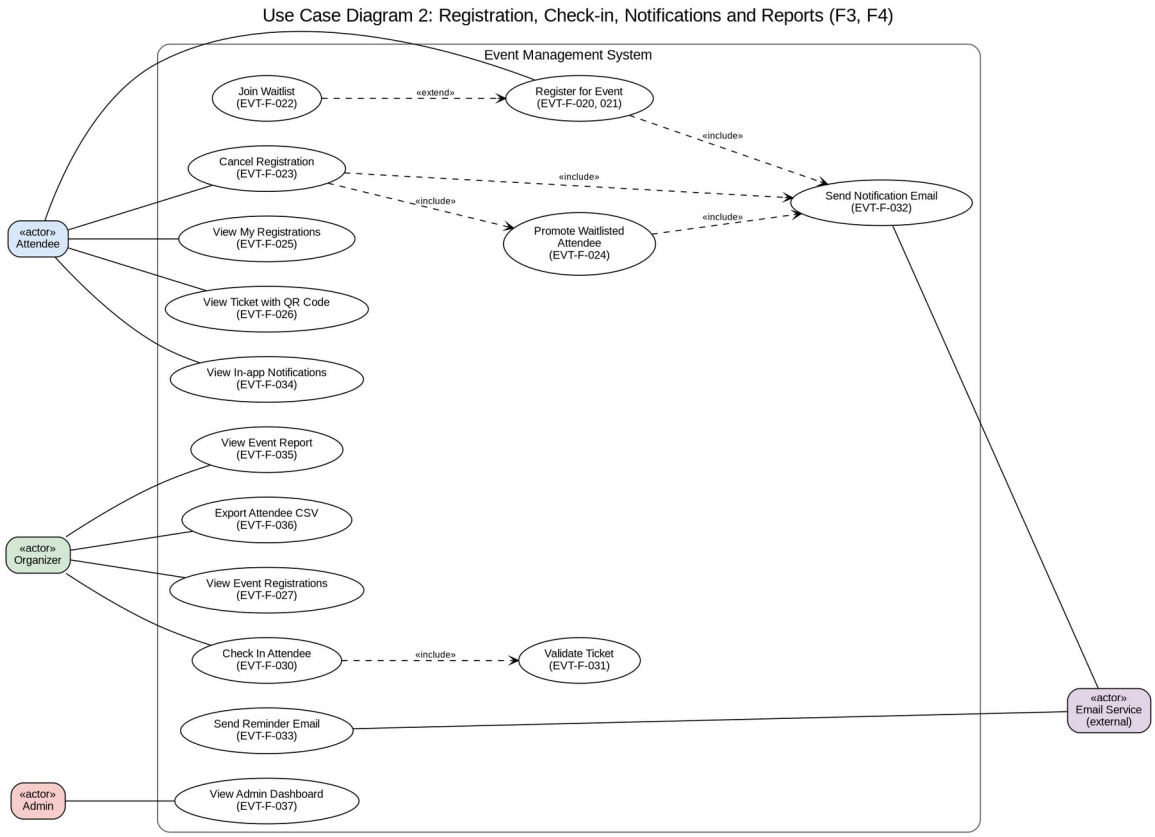


Figure 7.2: Use case diagram 2, registration, check-in and reports (F3, F4)

8. Requirements Traceability Matrix (RTM)

Each requirement is traced to its SRS section, its design in the SAD (component, API endpoint and sequence diagram), the code module that implements it, the test cases in the Test Plan, and the result of running those test cases on the final Sprint 2 build (10-11-2026).

Status: P = all test cases passed; A = passed with an accepted deviation or note (see comments); N = not yet fully tested.

Req ID	Requirement short	Section ref	Design spec (SAD)	Module	Test case(s)	Status (N/P/A)	Comments
EVT-F-001	Register user account	4.1	Auth Module; POST /api/auth/register	backend/src/modules/auth	TC-AUTH-01, TC-AUTH-02, TC-AUTH-10	P	
EVT-F-002	Log in with email and password	4.1	Auth Module; POST /api/auth/login; SD: SD-1	backend/src/modules/auth	TC-AUTH-03, TC-AUTH-09	P	
EVT-F-003	Lock account after repeated failed logins	4.1	Auth Module, Audit Logger; POST /api/auth/login; SD: SD-1	backend/src/modules/auth	TC-AUTH-04	P	
EVT-F-004	Log out and invalidate session	4.1	Auth Module, Security Middleware; POST /api/auth/logout	backend/src/modules/auth	TC-AUTH-05	P	
EVT-F-005	Role-based access control	4.1	Security Middleware (RBAC); All protected endpoints	backend/src/middleware/rbac.js	TC-AUTH-06, TC-EVT-10	P	
EVT-F-006	Admin user and role management	4.1	Auth Module; GET /api/users, PATCH /api/users/:id/role, PATCH /api/users/:id/status	backend/src/modules/auth	TC-AUTH-07	P	
EVT-F-007	Profile and password management	4.1	Auth Module; GET and PUT /api/users/me, PUT /api/users/me/password	backend/src/modules/auth	TC-AUTH-08	P	
EVT-F-010	Create event	4.2	Event Module; POST /api/events	backend/src/modules/events	TC-EVT-01, TC-EVT-10	P	
EVT-F-011	Validate event data	4.2	Event Module, Security Middleware (validation); POST /api/events, PUT /api/events/:id	backend/src/modules/events	TC-EVT-02, TC-EVT-03	P	
EVT-F-012	Edit own event	4.2	Event Module; PUT /api/events/:id	backend/src/modules/events	TC-EVT-04	P	
EVT-F-013	Publish draft event	4.2	Event Module; POST /api/events/:id/publish	backend/src/modules/events	TC-EVT-05	P	
EVT-F-014	Cancel event and its registrations	4.2	Event Module, Registration Module, Notification Module; POST /api/events/:id/cancel; SD: SD-2	backend/src/modules/events	TC-EVT-06	P	
EVT-F-015	Manage event categories	4.2	Event Module; /api/categories endpoints	backend/src/modules/events	TC-EVT-07	P	
EVT-F-016	Browse, search and filter events	4.2	Event Module; GET /api/events, GET /api/events/:id	backend/src/modules/events	TC-EVT-08, TC-EVT-09	A	TC-EVT-09: only upcoming published events are listed; cancelled events are not shown with a label (deviation)
EVT-F-020	Register for an event	4.3	Registration Module; POST /api/events/:id/registrations; SD: SD-3	backend/src/modules/registrations	TC-REG-01, TC-REG-02	P	

Req ID	Requirement short	Section ref	Design spec (SAD)	Module	Test case(s)	Status (N/P/A)	Comments
EVT-F-021	Prevent duplicate registrations	4.3	Registration Module; POST /api/events/:id/registrations; SD: SD-3	backend/src/modules/registrations	TC-REG-03	P	
EVT-F-022	Join waitlist when event is full	4.3	Registration Module; POST /api/events/:id/registrations; SD: SD-3	backend/src/modules/registrations	TC-REG-04	P	
EVT-F-023	Cancel registration	4.3	Registration Module; POST /api/registrations/:id/cancel	backend/src/modules/registrations	TC-REG-05, TC-REG-10	P	
EVT-F-024	Promote first waitlisted attendee	4.3	Registration Module, Notification Module; POST /api/registrations/:id/cancel	backend/src/modules/registrations	TC-REG-06	P	
EVT-F-025	View my registrations	4.3	Registration Module; GET /api/registrations/me	backend/src/modules/registrations	TC-REG-07	P	
EVT-F-026	Ticket with QR code	4.3	Registration Module (TicketService); GET /api/registrations/:id/ticket	backend/src/modules/registrations	TC-REG-08	P	
EVT-F-027	Organizer views event registrations	4.3	Registration Module; GET /api/events/:id/registrations	backend/src/modules/registrations	TC-REG-09	P	
EVT-F-030	Check in attendee by ticket ID	4.4	Check-in Module; POST /api/events/:id/checkins; SD: SD-4	backend/src/modules/checkin	TC-CHK-01	P	
EVT-F-031	Reject invalid tickets at check-in	4.4	Check-in Module; POST /api/events/:id/checkins; SD: SD-4	backend/src/modules/checkin	TC-CHK-02, TC-CHK-03	P	
EVT-F-032	Registration status emails	4.4	Notification Module; Internal calls from Event and Registration modules; SD: SD-2, SD-3	backend/src/modules/notifications	TC-CHK-04	P	
EVT-F-033	24-hour reminder emails	4.4	Notification Module, Scheduler; Internal scheduled job	backend/src/modules/notifications (scheduler)	TC-CHK-05	P	
EVT-F-034	In-app notifications	4.4	Notification Module; GET /api/notifications, PATCH /api/notifications/:id/read	backend/src/modules/notifications	TC-CHK-06	P	
EVT-F-035	Event attendance report	4.4	Reporting Module; GET /api/events/:id/report	backend/src/modules/reports	TC-CHK-07, TC-CHK-10	P	
EVT-F-036	Export attendee list as CSV	4.4	Reporting Module; GET /api/events/:id/attendees.csv	backend/src/modules/reports	TC-CHK-08	P	
EVT-F-037	Admin dashboard	4.4	Reporting Module; GET /api/admin/dashboard	backend/src/modules/reports	TC-CHK-09	P	
EVT-NF-001	Performance: 95% within 2 s	5	All modules, database indexes; All endpoints	backend/scripts/load-test.js, database indexes	TC-NFR-01	P	Team load-test script used instead of JMeter
EVT-NF-002	Availability: 99% monthly uptime	5	Deployment (Docker), health check; GET /health	health module, docker-compose.yml	TC-NFR-02	A	Measured over a short test period; a month needs a deployed server
EVT-NF-003	Usability and responsive layout	5	React SPA	frontend/src (pages, styles.css)	TC-NFR-03	P	
EVT-NF-004	No overbooking under concurrency	5	Registration Module (transaction with row lock); POST /api/events/:id/registrations; SD: SD-3	backend/src/modules/registrations	TC-NFR-04	P	
EVT-NF-005	70% unit test coverage and quality gate	5	CI pipeline, test suites, SonarQube	backend/package.json (coverage gate), CI	TC-NFR-05	N	Coverage 86.8% passes; SonarQube run pending
EVT-NF-006	Browser compatibility	5	React SPA	frontend/vite.config.js	TC-NFR-06	N	Chrome passes; Firefox and

Req ID	Requirement short	Section ref	Design spec (SAD)	Module	Test case(s)	Status (N/P/A)	Comments
							Edge pending
EVT-SR-001	Passwords stored as bcrypt hashes	5.1.2	Auth Module (bcrypt); POST /api/auth/register, POST /api/auth/login; SD: SD-1	backend/src/modules/auth	TC-SEC-01	P	
EVT-SR-002	Signed JWT sessions, 60 minutes	5.1.2	Security Middleware (JWT); All protected endpoints; SD: SD-1	backend/src/middleware/auth.js	TC-SEC-02	P	
EVT-SR-003	Input validation and parameterized queries	5.1.2	Security Middleware (validation), Repositories; All endpoints that accept input	validation files, utils/sanitize.js	TC-SEC-03	P	
EVT-SR-004	HTTPS with TLS 1.2 or higher	5.1.2	Deployment (HTTPS reverse proxy); All endpoints	frontend/nginx.conf, Dockerfile	TC-SEC-04	P	Self-signed certificate for local use (deviation)
EVT-SR-005	Ownership checks	5.1.2	Security Middleware, Registration and Event modules (ownership checks); Registration and event endpoints; SD: SD-2, SD-4	service ownership checks	TC-SEC-05	P	
EVT-SR-006	Audit logging of security events	5.1.2	Audit Logger (used by the Auth, Event and Check-in modules); Login, role change, event cancel, check-in; SD: SD-1, SD-2, SD-4	backend/src/utils/auditLogger.js	TC-SEC-06	P	
EVT-SR-007	Rate limiting	5.1.2	Security Middleware (rate limiter); All endpoints, strictest on login	backend/src/middleware/rateLimiter.js	TC-SEC-07	P	

Summary: 43 requirements, each traced to at least one test case. Status P: 39, A: 2, N: 2. No requirement failed its tests.

8.1 Implementation deviations

The requirements did not change during implementation. Where the implementation differs in detail from this SRS or from the SAD, the difference and the reason are recorded in the sprint reviews and summarized in SAD section 4.6.1. The ones that touch requirements in this SRS:

- EVT-F-016: search lists only upcoming published events; a cancelled event is not listed, its own page shows "Cancelled".
- EVT-F-033: the reminder job runs every minute and reminds each Confirmed attendee once when the event is within 24 hours, so late registrations also get a reminder.
- EVT-F-035: "Confirmed" in the report includes attendees already checked in, so the attendance rate is checked-in divided by all seat holders.
- EVT-F-012, F-014, F-027, F-030, F-035, F-036: admins may also perform these organizer actions.
- EVT-SR-004: HTTPS uses a self-signed certificate for local use (TLS 1.2 and 1.3 only).
- EVT-F-032: an extra email is sent when an attendee joins the waitlist.